



RAMCO INSTITUTE OF TECHNOLOGY

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Department of Computer Science and Engineering Academic Year 2025 – 2026 (Odd Semester)

Degree, Semester & Branch : III Semester B.E CSE B
Course Code & Title : CS3352 & Foundations of Data Science
Name of the Faculty member : Mrs. P.Sabeena Burvin, AP / CSE

Innovative Practice Description

- **Unit / Topic:** Unit III / Standard error of estimate
- **Course Outcome:** CO3
- **Topic Learning Outcome:** TLO 1,2,3,6,7,8,9,10,11
- **Activity Chosen:** Learning by Teaching
- **Date of Implementation:** 18.09.2025
- **Justification:**

In contemporary statistics education, learners often struggle with abstract concepts such as the **Standard Error of Estimate (SEE)** is a measure of how closely data points cluster around a regression line. To promote deeper understanding, this innovative-teaching project implemented the **Learning by Teaching (LbT)** method, where students take on the role of teachers to explain concepts to peers. The aim was to improve comprehension, retention, and communication skills through active knowledge construction.

Time Allotted for the Activity: 30 Minutes

- **Details of the Implementation:**

Objectives:

- 1.Strengthen conceptual understanding of SEE and its role in regression analysis.
- 2.Develop students' ability to interpret statistical results in practical scenarios.
- 3.Enhance communication, collaboration, and problem-solving abilities through peer teaching.
- 4.Encourage self-directed learning and higher-order thinking.

Key Concepts Learned:

- They **organize information more deeply**,
- Identify gaps in their own understanding,
- Engage in **metacognitive reflection**, and
- Learn to communicate clearly.

Standard Error of Estimate (SEE)

SEE quantifies prediction accuracy in regression analysis.

$$SEE = \sqrt{\sum (Y - \hat{Y})^2 / n - 2}$$

A lower SEE indicates a better fit between observed and predicted values. Students frequently misuse or misinterpret SEE—making it ideal for LbT-based learning.

Preparation Phase:

- Students were divided into small “teaching groups.”
- Each group researched SEE: definition, formula, application, and interpretation.
- Groups prepared **mini-lessons** (15 minutes) including examples, analogies, and simple datasets.

Teaching Phase:

Each group taught their peers using:

- Visual aids (graphs of regression lines and residuals)
- Step-by-step SEE calculations
- Real-life contexts (predicting housing prices, exam scores, etc.)

Observed Outcomes:

- Improved accuracy in calculating SEE.
- Students demonstrated clearer distinction between **standard error**, **standard deviation**, and **SEE**.
- Peer explanations helped demystify the formula and its practical meaning.

Learning by Teaching approach proved to be an effective innovative strategy for teaching the Standard Error of Estimate. Students not only gained stronger conceptual understanding but also improved communication and collaboration skills. By shifting responsibility to learners, the method transformed statistical learning from passive reception to active knowledge construction

- **CO – PO / PSO mapping:**

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CO	PO1	PO2	PO9	PO12	PSO1	PSO3
CO1	2	2	3	2	2	2

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	PO1	PO2	PO9	PO12	PSO1	PSO3
	2	2	3	2	2	2
Justification for correlation	Apply the knowledge of mathematical formulation for Standard error of estimate	Identify the appropriate methods to calculate the Standard error of estimate	Individually identify the problem and finding solutions	Communicate / share the ideas with other students	Deliver efficient document and solve the problem step by step	Recognize the need for Standard error of estimate in correlation problems in data analysis

- **Images / Screenshot of the practice:**



Figure 1: Definition and Formulas presented by Kamalee



Figure 2: Standard Error estimation calculation presented by Kavitha

Reflective Critique:

❖ *Feedback of practice from students and other stakeholders:*

- Presenting the seminar helped to understand the topic more deeply as it had to explain it to others.
- Students found the seminar activity to be a valuable learning experience that enhanced their understanding, communication skills, and confidence

❖ *Benefit of the practice:* (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)

- By involving students in teaching and discussions, the seminar promotes active learning, making the educational experience more dynamic and interactive.
- Students spent their time in self-learning.
- The seminar provides an opportunity for students to receive feedback from both peers and instructors, helping them identify strengths and areas for improvement.

❖ *Challenges faced in implementation:*

- Due to shyness, some students hesitate to teach.
- Keeping all students engaged throughout the seminar can be challenging, especially if presentations are lengthy or not interactive.

References:

1. <https://effectiviology.com/protege-effect-learn-by-teaching/>
2. https://www.researchgate.net/publication/351905566_Impact_of_Seminars_on_Student_Soft_Skills_Development
3. https://www.ritrjpm.ac.in/images/computer-science/2022-2023/5_VA_CS3362_Learning%20by%20teaching.pdf
4. https://www.ritrjpm.ac.in/images/computer-science/2022-2023/Unit_5_Learning_by_Teaching.pdf

